

HDOC DAY -19

QUESTION:

The algorithm for calculating the Semester grade point average (SGPA) and Cumulative Grade Point Average (CGPA) for “m” course is given below. Each semester has “n” course where ‘n’ may vary between semesters.

Assume that the student has passed all courses. Prepare a **test-case table covering valid, invalid, and boundary inputs**; write the **C program based on the given algorithm**; compile and execute it; and test it against the prepared test cases.

Aim:

To implement a C program for calculating **SGPA (Semester Grade Point Average)** and **CGPA (Cumulative Grade Point Average)** for multiple semesters using **nested loops and decision-making constructs**.

Algorithm

1. **Start.**
2. Define the variables and their types:
global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC,
cumulative_TC, cumulative_TGC as integers;
grade as a character;
SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get m (number of semesters).
6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set cumulative values:
cumulative_TC = 0
cumulative_TGC = 0
8. Repeat the following steps for m semesters:

8.1 Set semester totals:

semester_TC = 0
semester_TGC = 0

8.2 Get n (number of courses).

8.3 Validate n: If n is not positive, display an error message and request it again until a positive value is entered.

8.4 Repeat the following for n courses:

- **8.4.1 Get code.**

- **8.4.2** Validate code. It must be positive.
- **8.4.3** Get credit.
- **8.4.4** Validate credit. It must be positive.
- **8.4.5** Get grade.
- **8.4.6** Validate grade. Accept only O, A, B, C, D, E, P. If invalid, request it again.
- **8.4.7** Find GP according to the grade:

$$O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4$$
- **8.4.8** Calculate semester total grade credits:

$$\text{semester_TGC} = \text{semester_TGC} + (\text{GP} \times \text{credit})$$
- **8.4.9** Calculate semester total credits:

$$\text{semester_TC} = \text{semester_TC} + \text{credit}$$

8.5 Calculate SGPA:

$$\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$$

8.6 Display semester and SGPA.

8.7 Add semester grade credits to cumulative grade credits:

$$\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$$

8.8 Add semester credits to cumulative credits:

$$\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$$

9. Calculate CGPA:

$$\text{CGPA} = \text{cumulative_TGC} / \text{cumulative_TC}$$

10. Display global_id, m, cumulative_TC, and CGPA.

11. Stop

Test Case	Input / Condition	Expected Result
1	Valid Global ID, semesters, courses, credits and grades	SGPA and CGPA calculated successfully
2	global_id = 0 or negative	Display Invalid Global ID and ask again
3	m = 0 or negative	Display Invalid number of semester and ask again
4	n = 0 or negative	Display Invalid number of course and ask again
5	Course code ≤ 0	Display Invalid course code and ask again
6	Credit ≤ 0	Display Invalid credit and ask again
7	Grade other than O, A, B, C, D, E, P	Display Invalid Grade and ask again
8	m = 1, n = 1, valid course details	Boundry case: Calculate SGPA and CGPA for one semester and one course

>Write a Code C:

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int global_id, m, n;
6     int sem, course;
7     int code, credit, GP;
8     int sem_TC, sem_TGC;
9     int total_TC = 0, total_TGC = 0;
10    char grade;
11    float SGPA, CGPA;
12
13    /* Global ID */
14    do
15    {
16        printf("Enter Global ID: ");
17        scanf("%d", &global_id);
18
19        if (global_id <= 0)
20            printf("Invalid Global ID. Enter again.\n");
21    } while (global_id <= 0);
22
23    /* Number of semesters */
24    do
25    {
26        printf("Enter number of semesters: ");
27        scanf("%d", &m);
28
29        if (m <= 0)
30            printf("Invalid number of semesters. Enter again.\n");
31    } while (m <= 0);
32
33    /* Semester loop */
34    for (sem = 1; sem <= m; sem++)
35    {
36        sem_TC = 0;
37        sem_TGC = 0;
38
39        printf("\nSemester %d\n", sem);
40
41        /* Number of courses */
42        do
43        {
44            printf("Enter number of courses: ");
45            scanf("%d", &n);
46
47            if (n <= 0)
48                printf("Invalid number of courses. Enter again.\n");
49        } while (n <= 0);
50    }
51
52    /* Calculate SGPA and CGPA */
53    for (sem = 1; sem <= m; sem++)
54    {
55        for (course = 1; course <= n; course++)
56        {
57            printf("Enter course code: ");
58            scanf("%d", &code);
59
60            if (code <= 0)
61                printf("Invalid course code. Enter again.\n");
62
63            printf("Enter credit: ");
64            scanf("%d", &credit);
65
66            if (credit <= 0)
67                printf("Invalid credit. Enter again.\n");
68
69            printf("Enter grade: ");
70            scanf("%c", &grade);
71
72            if (grade != 'O' & grade != 'A' & grade != 'B' & grade != 'C' & grade != 'D' & grade != 'E' & grade != 'P')
73                printf("Invalid grade. Enter again.\n");
74
75            GP = (code * credit);
76            sem_TC += GP;
77            sem_TGC += credit;
78
79            printf("Course %d, Credit %d, Grade %c, GP %d\n", course, credit, grade, GP);
80        }
81
82        SGPA = sem_TC / sem_TGC;
83        total_TC += sem_TC;
84        total_TGC += sem_TGC;
85    }
86
87    CGPA = total_TC / total_TGC;
88
89    printf("SGPA: %.2f\n", SGPA);
90    printf("CGPA: %.2f\n", CGPA);
91
92    return 0;
93 }

```

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printf("Invalid number of courses. Enter again.\n");
50
51 } while (n <= 0);
52
53 /* Course loop */
54 for (course = 1; course <= n; course++)
55 {
56     printf("\nCourse %d\n", course);
57
58     /* Course code */
59     do
60     {
61         printf("Enter course code: ");
62         scanf("%d", &code);
63
64         if (code <= 0)
65             printf("Invalid course code. Enter again.\n");
66     } while (code <= 0);
67
68     /* Credit */
69     do
70     {
71         printf("Enter credit: ");
72         scanf("%d", &credit);
73
74         if (credit <= 0)
75             printf("Invalid credit. Enter again.\n");
76     } while (credit <= 0);
77
78     /* Grade */
79     do
80     {
81         printf("Enter grade (O/A/B/C/D/E/P): ");
82         scanf("%c", &grade);
83
84         switch (grade)
85         {
86             case 'O':
87                 GP = 10;
88                 break;
89
90             case 'A':
91                 GP = 9;
92                 break;
93
94             case 'B':
95                 GP = 8;
96                 break;
97
98             case 'C':
99                 GP = 7;
100                 break;
101
102             case 'D':
103                 GP = 6;
104                 break;
105
106             case 'E':
107                 GP = 5;
108                 break;
109
110             case 'P':
111                 GP = 4;
112                 break;
113
114             default:
115                 GP = -1;
116                 printf("Invalid grade. Enter again.\n");
117         }
118     } while (GP == -1);
119
120     sem_TGC = sem_TGC + (GP * credit);
121     sem_TC = sem_TC + credit;
122 }
123
124 /* Calculate SGPA */
125 SGPA = (float)sem_TGC / sem_TC;
126 printf("\nSGPA of Semester %d = %.2f\n", sem, SGPA);
127
128 total_TGC = total_TGC + sem_TGC;
129 total_TC = total_TC + sem_TC;
130 }
131
132 /* Calculate CGPA */
133 CGPA = (float)total_TGC / total_TC;
134
135 printf("\nGlobal ID = %d\n", global_id);
136 printf("Number of Semesters = %d\n", m);
137 printf("Cumulative Credits = %d\n", total_TC);
138 printf("CGPA = %.2f\n", CGPA);
139
140 return 0;
141 }
```

```
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98 GP = 8;
99 break;
100
101 case 'C':
102     GP = 7;
103     break;
104
105 case 'D':
106     GP = 6;
107     break;
108
109 case 'E':
110     GP = 5;
111     break;
112
113 case 'P':
114     GP = 4;
115     break;
116
117 default:
118     GP = -1;
119     printf("Invalid grade. Enter again.\n");
120 }
121 } while (GP == -1);
122
123 sem_TGC = sem_TGC + (GP * credit);
124 sem_TC = sem_TC + credit;
125 }
126
127 /* Calculate SGPA */
128 SGPA = (float)sem_TGC / sem_TC;
129 printf("\nSGPA of Semester %d = %.2f\n", sem, SGPA);
130
131 total_TGC = total_TGC + sem_TGC;
132 total_TC = total_TC + sem_TC;
133 }
134
135 /* Calculate CGPA */
136 CGPA = (float)total_TGC / total_TC;
137
138 printf("\nGlobal ID = %d\n", global_id);
139 printf("Number of Semesters = %d\n", m);
140 printf("Cumulative Credits = %d\n", total_TC);
141 printf("CGPA = %.2f\n", CGPA);
142
143 return 0;
144 }
```

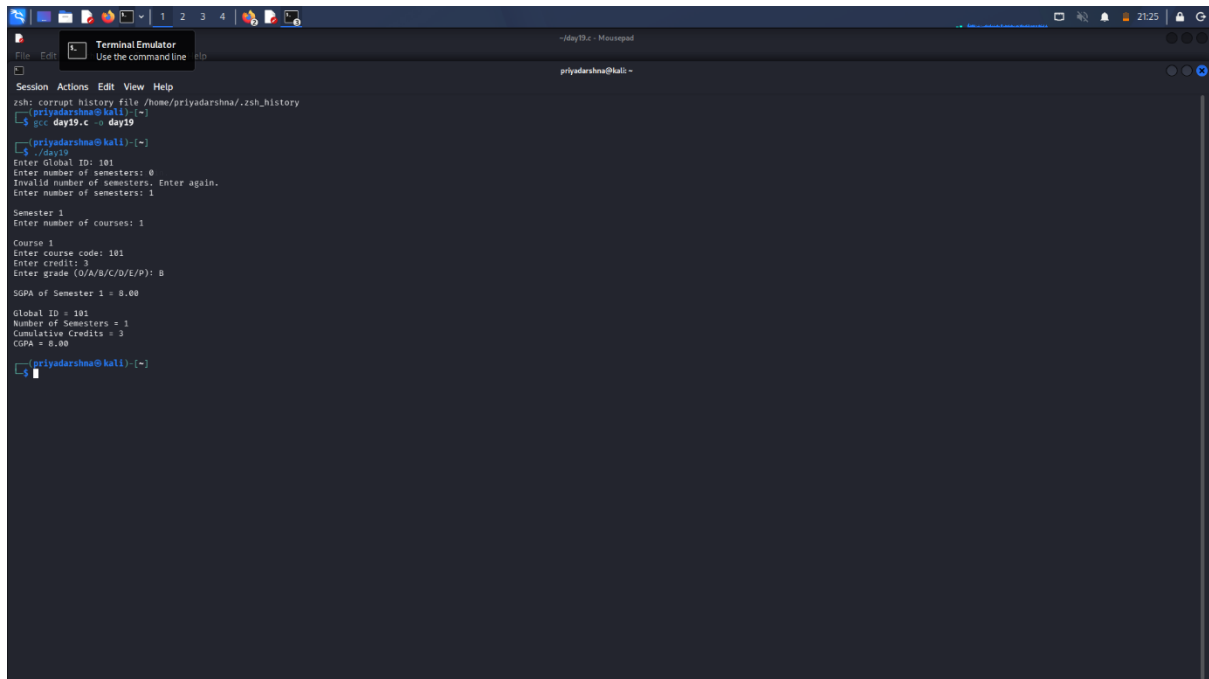
>TEST CASE 1:

```
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priyadarshna@kali ~
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali] ~
$ gcc day19.c -o day19
[priyadarshna@kali] ~
$ ./day19
Enter Global ID: 101
Enter number of semesters: 1
Semester 1
Enter number of courses: 2
Course 1
Enter course code: 101
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): 0
Course 2
Enter course code: 102
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): A
SGPA of Semester 1 = 9.57
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 7
CGPA = 9.57
[priyadarshna@kali] ~
$
```

>TEST CASE 2:

```
File Edit Search View Document Help
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priyadarshna@kali ~
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali] ~
$ gcc day19.c -o day19
[priyadarshna@kali] ~
$ ./day19
Enter Global ID: -5
Invalid Global ID. Enter again.
Enter Global ID: 101
Enter number of semesters: 1
Semester 1
Enter number of courses: 1
Course 1
Enter course code: 101
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): A
SGPA of Semester 1 = 9.00
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 4
CGPA = 9.00
[priyadarshna@kali] ~
$
```

>TEST CASE 3:



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day19.c -o day19
(priyadarshna@kali) ~
$ ./day19
Enter Global ID: 101
Enter Number of semesters: 0
Invalid number of semesters. Enter again.
Enter number of semesters: 1

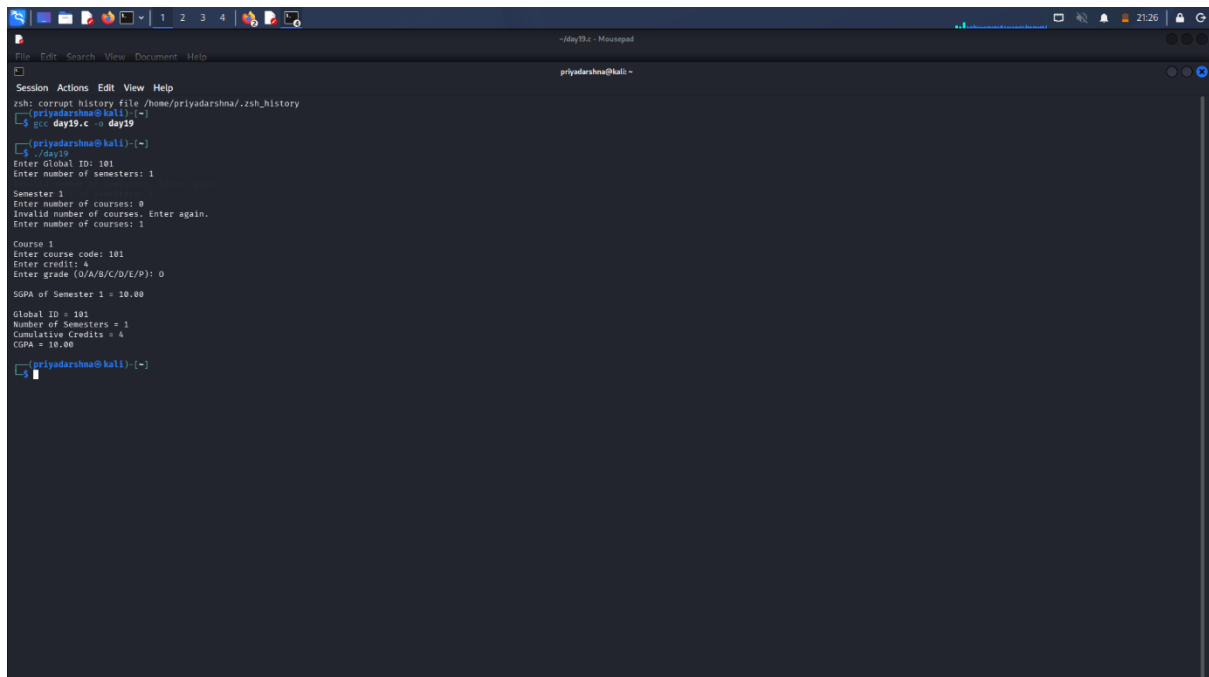
Semester 1
Enter number of courses: 1

Course 1
Enter course code: 101
Enter Credit: 3
Enter grade (O/A/B/C/D/E/P): B

SGPA of Semester 1 = 8.00

Global ID = 101
Number of Semesters = 1
Cumulative Credits = 3
CGPA = 8.00
(priyadarshna@kali) ~
$
```

>TEST CASE 4:



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day19.c -o day19
(priyadarshna@kali) ~
$ ./day19
Enter Global ID: 101
Enter number of semesters: 1

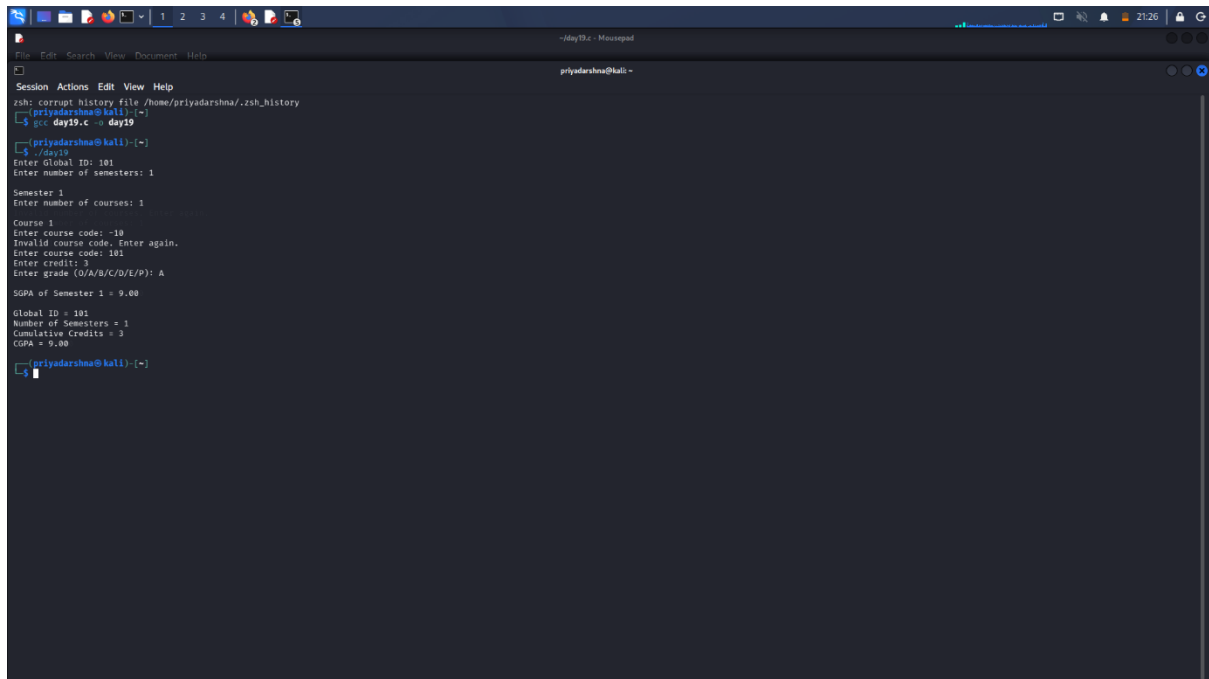
Semester 1
Enter number of courses: 0
Invalid number of courses. Enter again.
Enter number of courses: 1

Course 1
Enter course code: 101
Enter Credit: 4
Enter grade (O/A/B/C/D/E/P): O

SGPA of Semester 1 = 10.00

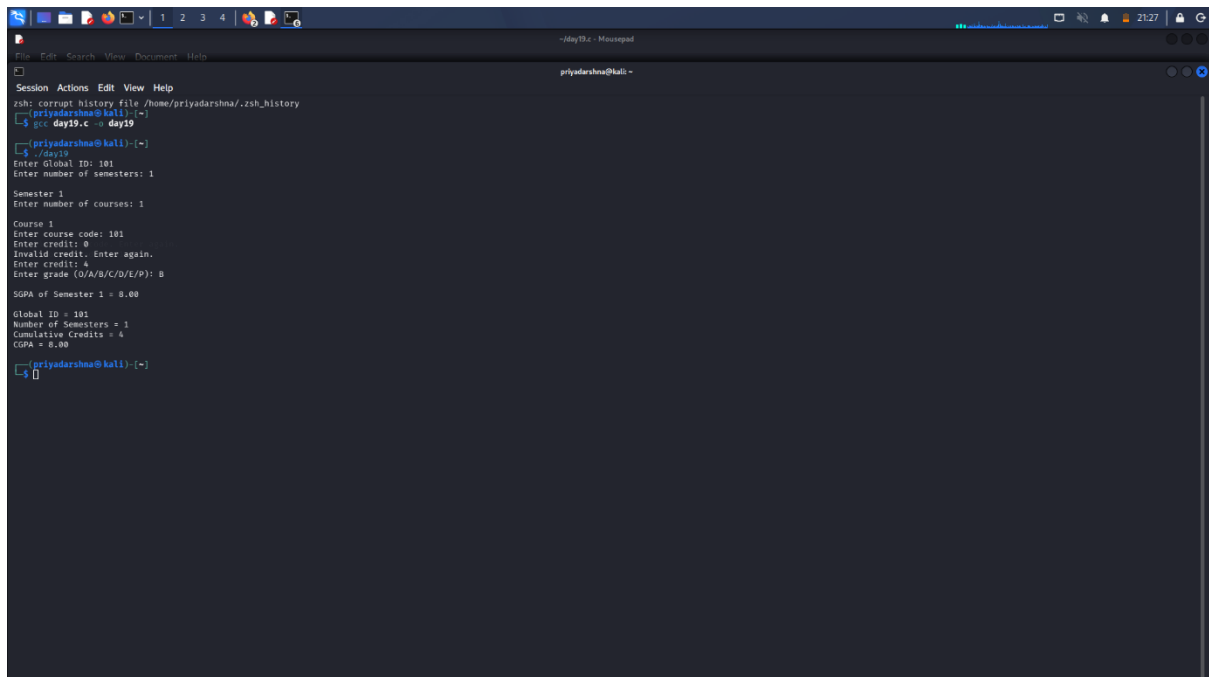
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 4
CGPA = 10.00
(priyadarshna@kali) ~
$
```

>TEST CASE 5:



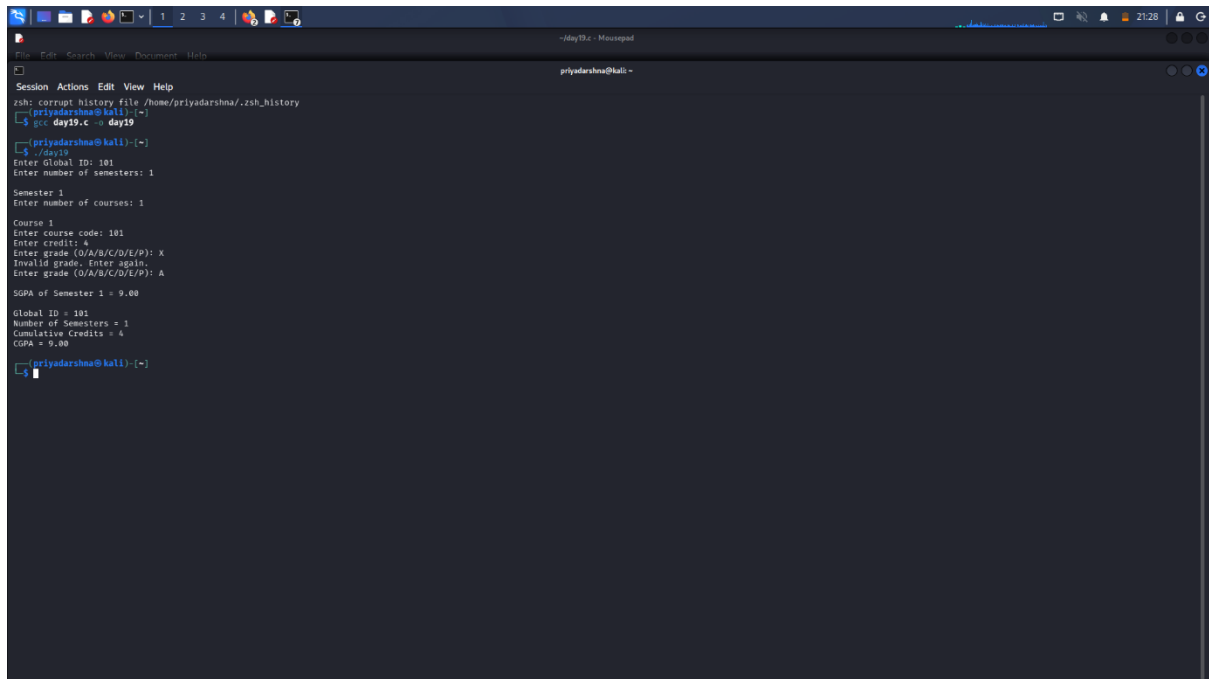
```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day19.c -o day19
(priyadarshna@kali) ~
$ ./day19
Enter Global ID: 101
Enter number of semesters: 1
Semester 1
Enter number of courses: 1
Course 1
Enter course code: -10
Invalid course code. Enter again.
Enter course code: 101
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): A
SGPA of Semester 1 = 9.00
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 3
CGPA = 9.00
(priyadarshna@kali) ~
$
```

>TEST CASE 6:



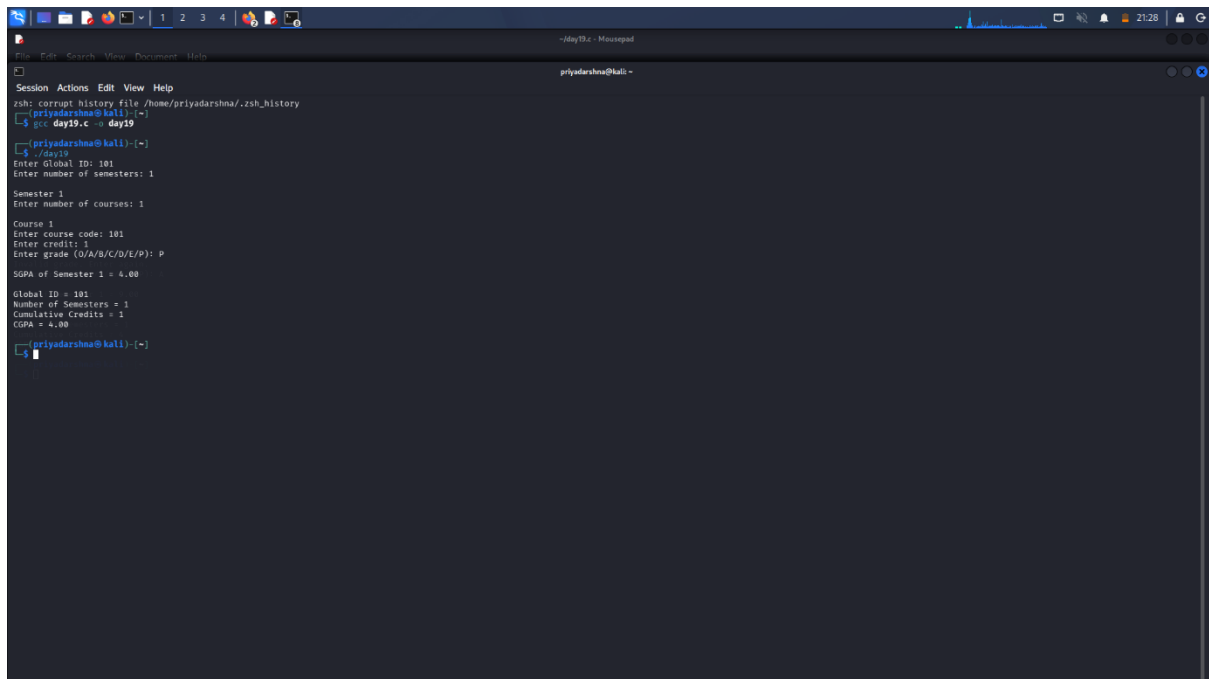
```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day19.c -o day19
(priyadarshna@kali) ~
$ ./day19
Enter Global ID: 101
Enter number of semesters: 1
Semester 1
Enter number of courses: 1
Course 1
Enter course code: 101
Enter credit: 0
Invalid credit. Enter again.
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): B
SGPA of Semester 1 = 8.00
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 4
CGPA = 8.00
(priyadarshna@kali) ~
$
```

>TEST CASE 7:



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day19.c -o day19
(priyadarshna@kali) ~
$ ./day19
Enter Global ID: 101
Enter number of semesters: 1
Semester 1
Enter number of courses: 1
Course 1
Enter course code: 101
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): X
Invalid grade. Enter again.
Enter grade (O/A/B/C/D/E/P): A
SGPA of Semester 1 = 9.00
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 4
CGPA = 9.00
(priyadarshna@kali) ~
$
```

>TEST CASE 8:



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day19.c -o day19
(priyadarshna@kali) ~
$ ./day19
Enter Global ID: 101
Enter number of semesters: 1
Semester 1
Enter number of courses: 1
Course 1
Enter course code: 101
Enter credit: 1
Enter grade (O/A/B/C/D/E/P): P
SGPA of Semester 1 = 4.00
Global ID = 101
Number of Semesters = 1
Cumulative Credits = 1
CGPA = 4.00
(priyadarshna@kali) ~
$
```

>RESULT :

Thus, the C program to calculate **SGPA and CGPA for multiple semesters using nested loops and decision-making constructs** was implemented, compiled, executed, and tested successfully with valid, invalid, and boundary test cases.